Independence as a Modeling Choice: The Geometry of Tractarian World Models

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Abstract

The *Tractatus* asserts that states of affairs are mutually independent (TLP 1.21, TLP 2.061–2.062); the color-exclusion problem (TLP 6.3751; Wittgenstein 1929) is standardly told as this thesis's refutation and the proximate cause of the book's collapse. We retell the story model-theoretically, on a machine-checked foundation. Rather than asking whether independence is *true*, we treat it as a *modeling choice*: the free model is a distinguished point in a space of world models ordered by a refinement preorder, whose structure we develop — evaluation is preserved along refinements, tautologies and contradictions are inherited downward, the free model is the maximum and is unique up to refinement-isomorphism among models realizing every truth-value profile, and refinement is a lattice-like order captured exactly by image-profile sets, with joins, meets, and infinitary joins. Constraint classes carve this space into tiers. For Horn constraints (positive implications between states of affairs) we state an exact realizability boundary that renders TLP 2.061 precise. Our central new result concerns *exclusion* constraints $\neg(a \wedge b)$ — the formal shape of color exclusion: no Horn constraint set is refinement-equivalent to any nontrivial exclusion model. Wittgenstein's 1929 crisis thus becomes a separation theorem about the geometry of logical space. All results are formalized in Lean 4 and checked by its kernel: 63 declarations, no `sorry`s, axiom footprint limited to propositional extensionality, choice, and quotient soundness.

1 Introduction

The *Tractatus Logico-Philosophicus* builds its account of logic and representation on a striking structural postulate: the mutual independence of states of affairs.

> 1.21 Each item can be the case or not the case while everything else remains the same.
>
> 2.062 From the existence or non-existence of one state of affairs it is impossible to infer the existence or non-existence of another. [1]

Independence is what makes the truth-table picture of propositions work: if n states of affairs can obtain or fail in any combination, logical space has exactly 2^n points, every elementary proposition is logically neutral with respect to every other (TLP 4.211), and all necessity is truth-functional necessity.

The standard narrative is that this postulate destroyed the book. Ramsey's review already pressed the problem of color ascriptions [3]; Wittgenstein himself flagged the case in TLP 6.3751 — "the simultaneous presence of two colours at the same place in the visual field is . . . ruled out by the logical structure of colour" — and in "Some Remarks on Logical Form" finally conceded that attributions of degree exclude one another in a way his truth-functional apparatus cannot represent [2]. Hacker's verdict is representative: Wittgenstein's first philosophy "collapsed over its inability to solve one problem — colour exclusion" [4].

This paper retells the story in a different register. We do not ask whether the independence thesis is true, nor whether some choice of elementary propositions rescues it (that is Moss's project [5], engaged in §7). We treat independence as a *modeling choice*: one admissible answer — the maximally permissive one — to the question of which truth-value profiles over the states of affairs are realized by some possible world. Under this reframing the interesting object is not the thesis but the *space of alternatives to it*: the collection of world models, ordered by how much independence they retain, with the fully independent *free model* at the top. The color-exclusion problem then stops being an embarrassment to be explained away and becomes a structural datum: the constraint that broke the book lives in a provably different region of this space than the constraints the book's machinery can absorb.

Concretely, we develop the following picture (Figure 1). A *world model* over a type S of states of affairs is a nonempty family of worlds together with a relation saying which states obtain at which world; its *image profiles* are the truth-value patterns over S it realizes. A *refinement* $M \sqsubseteq M'$ is a profile-preserving map from M-worlds to M'-worlds: passing down the order adds constraints, passing up removes them. Within this order:

- (C1) **The spectrum.** Refinement is a preorder with the free model as maximum; evaluation of propositions is preserved along refinements, so tautologies and contradictions propagate from freer to more constrained models; and a proposition holds at *every* model of the spectrum iff it is a tautology of the free model. Logical truths are exactly the propositions insensitive to the modeling choice (§3).
- (C2) **Uniqueness of the free model.** Among models realizing every profile, the free model is unique up to refinement-isomorphism: full independence pins the model down (§3).
- (C3) **The lattice of logical space.** Refinement between models is equivalent to inclusion of image-profile sets, and the order carries joins, meets, and infinitary joins computed exactly by unions and intersections of image profiles. The "geometry" of Tractarian logical space is the powerset lattice of 2^S in light disguise (§4).
- (C4) **An exact independence boundary.** For Horn constraints — positive implications $a \to b$ between states — a truth-value profile is realizable iff it satisfies every clause. This gives TLP 2.061 a precise, checkable content: independence fails in a Horn model exactly at the profiles the clauses exclude, and nowhere else (§5).
- (C5) **Color exclusion transcends the Horn tier.** Exclusion constraints $\neg(a \wedge b)$ also admit an exact realizability characterization, but no Horn constraint set is refinement-equivalent to any nontrivial exclusion model. The witness is the *top world* at which every state obtains: positive implications can never rule it out, any exclusion clause must. Instantiated at two states "point p is red" and "point p is green", this is TLP 6.3751 rendered as a separation theorem (§6).

Every result above is formalized in Lean 4 [17] and checked by its kernel. The base formalization of worlds, propositions, and evaluation is that of our companion paper on saying and showing [12], which this paper cites but does not depend on philosophically; the spectrum, Horn, equivalence, and exclusion developments comprise 63 declarations (∼1,000 lines) with no `sorry`s and an axiom footprint of `propext`, `Classical.choice`, and `Quot.sound` only. Theorem statements in the paper are cross-referenced to Lean declarations in Appendix A; we display Lean statements where the formal shape is load-bearing and paraphrase elsewhere.

2 World Models and the Free Model

Fix a type S whose elements we read as *states of affairs* (*Sachverhalte*). The base notions are inherited from the companion formalization [12].

Definition 2.1 (World model)**.** *A* world model M *over* S *consists of a type* $M.W$ *of worlds, a relation* $\mathrm{holds}_M : M.W \to S \to \mathrm{Prop}$*, and a proof that* $M.W$ *is nonempty. The* profile *of a world* $w \in M.W$ *is the function* $s \mapsto \mathrm{holds}_M(w, s)$*, the pattern of states obtaining at* w*. (Lean: `WorldModel`.)*

Definition 2.2 (Free model)**.** *The* free model *over* S *takes worlds* $= (S \to \mathrm{Prop})$ *and* $\mathrm{holds}(w, s) = w(s)$*: a world just* is *a truth-value profile, and every profile occurs. (Lean: `freeModel`.)*

The free model is the Tractarian picture of TLP 1.21 and TLP 2.062 taken at face value: any combination of states obtaining and failing is a way the world could be. Propositions are built from atoms $s \in S$ by negation and conjunction and evaluated at a world of a model in the obvious way (Lean: `Proposition`, `evalM`); a proposition is a *tautology of* M when it holds at every M-world.

The property that will carry the independence thesis through the paper is:

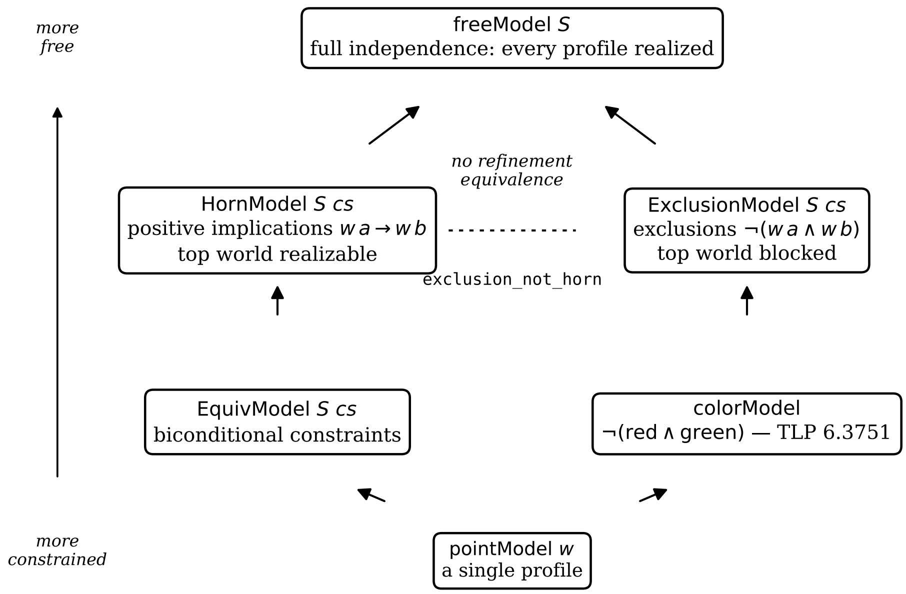

Figure 1: The world-model spectrum. Arrows are refinements (constraint removal); the free model is the maximum ((C1)). The Horn and exclusion tiers are refinement-incomparable: no Horn model shares a nontrivial exclusion model's image profiles (Theorem 6.5).

Definition 2.3 (Independent profles)**.** M has independent profiles *when every function* $S \to \mathrm{Prop}$ *is the profile of some* M*-world. (Lean:* `HasIndependentProfiles`*.)*

This is TLP 2.062 stated extensionally: nothing about which states obtain constrains which others do, because every combination is realized. The free model has independent profiles trivially (`freeModel_hasIndependentProfiles`); the question of the paper is what the landscape looks like when one gives this up.

3 The Refinement Spectrum

Definition 3.1 (Refinement)**.** M refines into M', *written* $M \sqsubseteq M'$, *when there is a map* $f : M.W \to M'.W$ *with* $\mathrm{holds}_M(w,s) \leftrightarrow \mathrm{holds}_{M'}(f(w),s)$ *for all* w, s*: every* M*-world's profile is realized in* M'*. (Lean:* `Refines`*.)*

Reading $\sqsubseteq$ upward is *constraint removal*: if $M \sqsubseteq M'$ then M' realizes at least the ways-the-world-could-be that M does. The following are elementary but jointly set the stage.

Theorem 3.2 (The spectrum)**.** (i) $\sqsubseteq$ *is reflexive and transitive.* (ii) $M \sqsubseteq \mathrm{freeModel}$ *for every* M*: the free model is the maximum of the spectrum.* (iii) *Evaluation is preserved along refinements: if* f *witnesses* $M \sqsubseteq M'$ *then every proposition has the same truth value at* w *and at* $f(w)$*.* (iv) *Hence tautologies and contradictions are inherited* downward*: if* p *is a tautology (contradiction) of* M' *and* $M \sqsubseteq M'$*, then* p *is a tautology (contradiction) of* M*. (Lean:* `refines_refl`*,* `refines_trans`*,* `refines_freeModel`*,* `refines_preserves_eval`*,* `tautology_pullback`*,* `contradiction_pullback`*.)*

Part (iv) is the formal cash value of the slogan that constraints *create* necessities: a model lower in the spectrum validates everything the freer model validates, plus possibly more. The converse direction — which propositions are validated *everywhere* — is answered exactly:

Theorem 3.3 (Spectrum-invariant validity)**.** *A proposition is a tautology of every world model over* S *iff it is a tautology of the free model. (Lean:* `spectrum_invariant_iff_freeModel_tautology`*; the forward direction factors through single-profile* point models*,* `pointModel_isTautology_iff`*.)*

The logical truths, in this setting, are precisely the propositions insensitive to the modeling choice. A Tractarian gloss: what holds in *logic* does not depend on how much independence the world exhibits — which is TLP 6.1's "the propositions of logic are tautologies" relativized to the whole spectrum rather than to logical space of the free kind.

Uniqueness gives the free model its privileged status among choices:

Theorem 3.4 (Uniqueness of the free model)**.** *If M has independent profiles, then M is refinement-isomorphic to the free model: there are refinements in both directions. Full independence determines the model up to the spectrum's own notion of equivalence. (Lean:* `freeModel_unique_refines_iso`*.)*

So the Tractarian choice is well-posed: "the model on which every combination of states is possible" picks out one point of the spectrum. The substantive question is what lies below it.

4 The Lattice of Logical Space

Refinement, defined via maps of worlds, is captured extensionally by a simpler invariant.

Definition 4.1 (Image profiles)**.** $\mathrm{Im}(M) \subseteq (S \to \mathrm{Prop})$ *is the set of profiles of M-worlds. (Lean:* `ImageProfiles`*.)*

Theorem 4.2 (Extensionality of refinement)**.** *$M \sqsubseteq M'$ iff $\mathrm{Im}(M) \subseteq \mathrm{Im}(M')$; and M, M' are refinement-equivalent (refinements both ways) iff $\mathrm{Im}(M) = \mathrm{Im}(M')$. Moreover $\mathrm{Im}(\mathrm{freeModel})$ is everything. (Lean:* `refines_iff_subset_imageProfiles`, `refinesEquiv_iff_image_eq`, `imageProfiles_freeModel`*.)*

Up to refinement equivalence, then, a world model over S *is* a nonempty set of profiles, and the spectrum is the powerset lattice of 2^S minus its bottom. The formalization equips it with the corresponding operations: a join model realizing the union of two models' profiles, a meet model realizing the intersection (when nonempty), and an infinitary join, each characterized by the expected universal property (Lean: `JoinModel`, `MeetModel`, `iJoinModel` with `refines_join_iff`, `refines_meet_iff`, `refines_iJoin_iff`).

This is the sense of "geometry" in our title, and it earns its keep twice over. First, it converts questions about world-models-up-to-equivalence into questions about *which sets of profiles a constraint class can carve out* — the form in which the separation theorem of §6 is proved. Second, it locates the Tractarian picture inside a familiar mathematical object: choosing independence is choosing the top of the lattice; every alternative model is a region of logical space; and TLP 3.42's dictum that "a proposition can determine only one place in logical space, but the whole of logical space must already be given by it" reads naturally as the observation that the ambient lattice is fixed by S alone, however the modeling choice restricts the realized region.

5 Constraint Tiers: Equivalence, Horn, and the Boundary of Independence

Arbitrary profile sets are unstructured; the Tractarian dialectic concerns constraints with an intelligible logical form. Two tiers from the formalization organize the well-behaved cases.

Definition 5.1 (Horn tier)**.** *For a list cs of pairs over S, the* Horn model $\mathrm{Horn}(S, cs)$ *has as worlds those profiles* $w : S \to \mathrm{Prop}$ *satisfying $w(a) \to w(b)$ for every $(a, b) \in cs$. (Lean:* `HornModel`*; each pair is a definite binary Horn clause.)*

Definition 5.2 (Equivalence tier)**.** $\mathrm{Equiv}(S, cs)$ *has as worlds the profiles satisfying $w(a) \leftrightarrow w(b)$ for every $(a, b) \in cs$. (Lean:* `EquivModel`*.)*

Biconditional constraints are the symmetric closure of implicational ones, and the formalization makes the containment official: every equivalence model is refinement-equivalent to the Horn model over the symmetrized clause list, and refines into the Horn model over the original one (Lean: `equivModel_iso_hornModel_symm`, `refines_equivModel_hornModel`). Both tiers break independence as soon as they say anything: a clause (a, b) with $a \neq b$ blocks the profile making a obtain and b fail (Lean: `hornModel_independence_fails`, `equivModel_independence_fails`).

What makes the Horn tier more than an example is that its incursion into independence can be characterized *exactly*.

Theorem 5.3 (Exact Horn realizability; the TLP 2.061 boundary)**.** *A Boolean valuation* $v : S \to \mathrm{Bool}$ *is the profile of some world of* $\mathrm{Horn}(S, cs)$ *iff* v *satisfies every clause: for all* $(a, b) \in cs$*, if* $v(a)$ *then* $v(b)$*. (Lean:* `horn_realizable_iff`*.)*

Read contrapositively, this says precisely which inferences "from the existence of one state of affairs to the existence of another" (TLP 2.062) a Horn regime licenses: exactly the clause-generated ones, and no others. Independence does not fail catastrophically or holistically; it fails at the excluded profiles and is retained everywhere else. The concrete case study in the formalization — a two-state weather model where "it rains" implies "the street is wet" — witnesses the boundary at minimal scale (Lean: `weatherModel_equiv_hornModel`, `weatherModel_horn_independence_fails`).

Theorem 5.3 is the paper's rendering of what is *right* in the Tractatus's treatment of constraint: where dependence has positive-implicational form, logical space contracts in a lawlike, fully-characterizable way. The next section shows that color exclusion is not of this form — provably.

6 Color Exclusion and the Limits of the Horn Tier

TLP 6.3751 locates the mutual exclusion of colors in "the logical structure of colour"; the 1929 paper concedes that such exclusions attach to attributions of degree quite generally and cannot be conjured out of truth-functional combination [2]. The formal shape of the datum is an *exclusion constraint*: two states that cannot both obtain.

Definition 6.1 (Exclusion tier)**.** $\mathrm{Excl}(S, cs)$ *has as worlds the profiles satisfying* $\neg(w(a) \wedge w(b))$ *for every* $(a, b) \in cs$*. (Lean:* `ExclusionModel`*.)*

The exclusion tier is as well-behaved, taken by itself, as the Horn tier:

Theorem 6.2 (Exact exclusion realizability)**.** $v : S \to \mathrm{Bool}$ *is the profile of some world of* $\mathrm{Excl}(S, cs)$ *iff no clause* $(a, b) \in cs$ *has* $v(a)$ *and* $v(b)$ *both true. (Lean:* `exclusion_realizable_iff`*.)*

Theorem 6.3 (Exclusion breaks independence)**.** *If* cs *is nonempty,* $\mathrm{Excl}(S, cs)$ *does not have independent profiles — with no side condition: even a degenerate self-clause* (a, a) *forbids* $w(a)$ *outright, so the all-true profile is always excluded. (Lean:* `exclusionModel_independence_fails`*; compare the Horn analog, which needs* $a \neq b$*.)*

The asymmetry flagged in Theorem 6.3 is the visible tip of a structural difference. At the top of every Horn model sits the world where everything obtains: a positive implication can never rule it out, since its consequent is true there. Exclusion clauses are defined by ruling it out.

Lemma 6.4 (The top world)**.** *For every clause list* cs*, the all-true profile lies in* $\mathrm{Im}(\mathrm{Horn}(S, cs))$*; for every* nonempty cs*, it does not lie in* $\mathrm{Im}(\mathrm{Excl}(S, cs))$*. (Lean:* `hornModel_allTrue_realizable`*,* `exclusionModel_allTrue_not_re`

Theorem 6.5 (Exclusion is not Horn)**.** *Let* cs *be nonempty. There is no clause list* ds *such that* $\mathrm{Horn}(S, ds)$ *is refinement-equivalent to* $\mathrm{Excl}(S, cs)$*. (Lean:* `exclusion_not_horn`*.)*

Proof sketch. By Theorem 4.2, refinement equivalence is equality of image-profile sets; by Lemma 6.4 the all-true profile separates any Horn image from any nontrivial exclusion image. The kernel-checked proof is nine lines. □

Remark 6.6. The top-world lemma is the nullary instance of a classical fact: Horn classes are closed under intersections of models (in the propositional case, pointwise conjunction of satisfying assignments), a property isolated by Horn in 1951 [15] following McKinsey's study of the clause class [14]. The empty intersection is the all-true assignment. Exclusion constraints fail closure under intersection in the most flagrant way available, and Theorem 6.5 is the Tractarian payoff of that failure. What our development adds to the classical picture is the refinement-theoretic phrasing — the two tiers are incomparable regions of the model spectrum, not merely distinct syntactic classes — and the mechanization.

The case study instantiates the general theorems at the historically loaded example. Let $S = \{\text{red}, \text{green}\}$, read as "point p in the visual field is red (green)", and let $\text{colorModel} = \text{Excl}(S, [(\text{red}, \text{green})])$.

Corollary 6.7 (TLP 6.3751, formalized)**.** colorModel *does not have independent profiles, and no Horn constraint list over S yields a model refinement-equivalent to it. (Lean:* `color_independence_fails`, `color_not_horn`*.)*

We propose reading Corollary 6.7 as a precise statement of what went wrong in 1929, one that improves on the collapse narrative in two ways. First, it separates two failures the narrative runs together. That color exclusion *breaks independence* (the first conjunct) is cheap: any substantive constraint does that, including the innocuous weather implication of §5. What distinguishes color exclusion is the second conjunct: it cannot be redescribed within the constraint class whose incursions into independence are exactly characterizable (Theorem 5.3) and closed under the operations that keep logical space lawlike. The 1929 crisis was not that the world exhibits dependence, but that it exhibits dependence of the wrong *shape*. Second, the reading makes the crisis *stable under reinterpretation*: Theorem 6.5 quantifies over all Horn clause lists, so no rewriting of the color facts in positive-implicational terms can dissolve it. The theorem does for the color-exclusion problem what Wittgenstein's own truth-table argument in the 1929 paper gestured at — showing the exclusion is not an artifact of notation — but with the quantifier over notations made explicit and discharged by the kernel.

7 Discussion

7.1 Button: cardinality versus structure

Button proves that the atomist picture tolerates exclusion problems iff logical space has cardinality a power of two, and reads the result as a reductio of Tractarian atomism given a continuum of color shades [6]. Our results are complementary but pull in a different direction: where Button's condition is global and cardinality-theoretic, the spectrum analysis is local and structural. Theorem 6.5 distinguishes exclusion from implication even over a two-element S, where cardinality considerations see nothing (both tiers carve out sets of profiles of unexceptional sizes). The moral we draw is that "how big is logical space?" undersells the atomist's predicament; the sharper question is "which regions of the profile lattice can constraints of a given logical form reach?" — and that question has exact answers (Theorems 5.3 and 6.2) that the cardinality frame cannot express.

7.2 Moss: re-atomization as a move within the spectrum

Moss argues the color-incompatibility problem is solvable: choose elementary propositions wisely (her construction uses vector-valued magnitudes) and the exclusions come out logical after all [5]. Our framework does not adjudicate whether her atoms are the right ones; it does clarify what kind of move she is making. Re-atomization replaces S by a new state-type S' and relocates the world model: the constrained region of the old lattice is re-presented as the free top of a new one. Nothing in our theorems obstructs this — they are stated for fixed S — but the spectrum makes the price explicit: the moves available to the atomist are exhausted by (i) changing S (Moss), or (ii) descending the lattice for fixed S, and Theorem 6.5 closes off the hope that descent of a logically tame (Horn) kind might suffice. The two strategies can then be evaluated on their merits rather than run together; the formal framework for comparing atomizations themselves (a map between spectra over different state-types) is future work, and §8 records it as such.

7.3 Combinatorialism

Armstrong's combinatorial theory of possibility adopts the "Tractarian thesis of Independence" for wholly distinct states of affairs, with possible worlds as recombinations [8], following Skyrms's Tractarian nominalism [7]. In our terms, combinatorialism is the decision to model possibility by the free model. The spectrum offers the combinatorialist a principled fallback under pressure from exclusion phenomena: rather than defending full recombination or abandoning the picture, one may occupy a named point lower in the lattice and know exactly what one has paid — which recombinations are lost (Theorems 5.3, 6.2) and which necessities are gained (Theorem 3.2(iv)). The uniqueness theorem (3.4) sharpens the standard observation that recombination determines the space of worlds: it does so up to exactly the equivalence the framework itself supplies.

7.4 Incompatibility as primitive

Evans and Berger's cathoristic logic takes incompatibility rather than negation as the primitive logical relation [13]. Their project is proof-theoretic and ours model-theoretic, but the underlying diagnosis converges: exclusion is not a derived phenomenon in a logic of positive combination. Theorem 6.5 can be read as the model-theoretic ground for that design decision — if exclusions were Horn-expressible, a logic with only positive primitives could simulate them, and by the theorem it cannot.

7.5 What the kernel settles

Every theorem cited above carries a machine-checked proof whose axiom footprint is `propext`, `Classical.choice`, `Quot.sound` — the standard classical base of Lean's mathematical library and nothing else; no `sorry`s remain. We follow the methodological line of the mechanization of Gödel's ontological argument [16]: where a philosophical dispute turns on *what follows from what*, mechanization relocates the dispute to its honest residue — whether the formal explication is faithful. For the present paper that residue is concentrated in two choices: reading TLP 2.062's independence as Definition 2.3 (realizability of every profile), and reading color exclusion as a binary exclusion clause (Definition 6.1). Both are defended in situ (§2, §6); a reader who rejects them knows exactly where to press, which is the methodological point. In the interest of full disclosure required by the method: statements and proofs of the exclusion-tier module were developed with an AI assistant and verified by the Lean kernel; the spectrum, Horn, and equivalence modules originate in the companion project [12], with machine-verified proofs throughout (see Acknowledgments).

8 Limitations

States of affairs are unstructured. S is a bare type: states have no inner composition of objects, so the account cannot yet express *why* red and green exclude (their sharing a determinable), only *that* an exclusion constraint is in force. A dependent-type treatment of objects, making independence claims sensitive to constituent-sharing, is the natural continuation (and was flagged as future work in the companion paper).

Binary clauses. Both the Horn and exclusion tiers use binary clause lists. n-ary Horn clauses and pairwise-exclusion families (a determinable with k determinates, e.g. a full color space) are unproblematic extensions of the definitions, but the theorems above are proved for the binary case; we expect Theorem 6.5's top-world argument to lift without incident, since it uses only the nullary closure instance.

Fixed atomization. The spectrum is indexed by a fixed S; comparing models across state-types — needed to fully engage Moss's re-atomization strategy — requires a notion of spectrum morphism we have not developed.

Explication risk. The kernel checks proofs, not faithfulness. Our TLP 2.061/TLP 2.062 and TLP 6.3751 readings are argued for but not beyond dispute; in particular, readers who take Tractarian elementary propositions to be constitutively independent (so that "constrained elementary propositions" is a contradiction in terms) will read the spectrum as charting successors to the Tractarian picture rather than variants of it. The theorems are indifferent to this choice of description.

No degree structure. The 1929 paper's own diagnosis involves attributions of degree and the joint syntax–phenomenology of measurement. Our exclusion clauses capture the resulting incompatibilities extensionally but say nothing about degrees as such.

9 Conclusion

Treating the Tractatus's independence thesis as a modeling choice replaces a binary dispute — independence: true or refuted? — with a structured space of positions, and the structure turns out to be theorem-rich. The free model sits at the top of a refinement lattice whose geometry is exactly the powerset of profiles;

spectrum-invariant validity coincides with free-model tautology; full independence determines the model uniquely up to the lattice's own equivalence. Constraint classes occupy the lattice as tiers with exactly characterizable reach, and the constraint that historically broke the book — color exclusion — provably outruns the reach of the positive-implicational tier. On this telling, the color-exclusion problem was never a mere counterexample awaiting a clever re-atomization or a regretful abandonment; it was the discovery, made without the benefit of the lattice in which to state it, that logical space has regions of essentially different shape. The machine-checked development makes that discovery a theorem, and the spectrum gives both the Tractarian and her critics named positions to occupy — with prices marked.

Acknowledgments

The base formalization of worlds, propositions, and evaluation, together with the spectrum, Horn-tier, and equivalence-tier modules, was developed in the companion project [12]; several of the ported modules' theorems were proved by Harmonic's Aristotle system from our statements, as acknowledged there. The exclusion-tier module (statements and proofs) was developed with Claude (Anthropic) in the course of preparing this paper. All proofs, whatever their origin, are checked by the Lean 4 kernel; the axiom footprint of every theorem cited here is `propext`, `Classical.choice`, `Quot.sound` only.

A Theorem Index

All declarations reside in the `tractatus` repository (`proofs/TractatusOntology{Horn,Equiv,Spectrum,Exclusion}.lean`) the build is warning-free under Lean 4.26 with no `sorry`s.

Paper	Lean declaration	Module
Thm 3.2(i)	`refines_refl`, `refines_trans`	Spectrum
Thm 3.2(ii)	`refines_freeModel`	Spectrum
Thm 3.2(iii)	`refines_preserves_eval`	Spectrum
Thm 3.2(iv)	`tautology_pullback`, `contradiction_pullback`	Spectrum
Thm 3.3	`spectrum_invariant_iff_freeModel_tautology`	Spectrum
Thm 3.4	`freeModel_unique_refines_iso`	Spectrum
Thm 4.2	`refines_iff_subset_imageProfiles`, `refinesEquiv_iff_image_eq`	Spectrum
(lattice ops)	`JoinModel`, `MeetModel`, `iJoinModel` + characterizations	Spectrum
Thm 5.3	`horn_realizable_iff`	Horn
(Horn indep. fails)	`hornModel_independence_fails`	Horn
(Equiv → Horn)	`equivModel_iso_hornModel_symm`, `refines_equivModel_hornModel`	Equiv
Thm 6.2	`exclusion_realizable_iff`	Exclusion
Thm 6.3	`exclusionModel_independence_fails`	Exclusion
Lem 6.4	`hornModel_allTrue_realizable`, `exclusionModel_allTrue_not_realizable`	Exclusion
Thm 6.5	`exclusion_not_horn`	Exclusion
Cor 6.7	`color_independence_fails`, `color_not_horn`	Exclusion

References

[1] Wittgenstein, Ludwig. *Tractatus Logico-Philosophicus*. Translated by D. F. Pears and B. F. McGuinness. London: Routledge & Kegan Paul, 1961. First published 1921 as "Logisch-philosophische Abhandlung".

[2] Wittgenstein, Ludwig. "Some Remarks on Logical Form." *Proceedings of the Aristotelian Society*, Supplementary Volume 9 (1929): 162–171.

[3] Ramsey, Frank P. "Critical Notice of L. Wittgenstein's *Tractatus Logico-Philosophicus*." *Mind* 32, no. 128 (1923): 465–478.

[4] Hacker, P. M. S. *Insight and Illusion: Themes in the Philosophy of Wittgenstein*. Revised edition. Oxford: Clarendon Press, 1986.

[5] Moss, Sarah. "Solving the Color Incompatibility Problem." *Journal of Philosophical Logic* 41, no. 5 (2012): 841–851. doi:10.1007/s10992-011-9193-3.

[6] Button, Tim. "Exclusion Problems and the Cardinality of Logical Space." *Journal of Philosophical Logic* 46, no. 6 (2017): 611–623. doi:10.1007/s10992-016-9412-z.

[7] Skyrms, Brian. "Tractarian Nominalism." *Philosophical Studies* 40, no. 2 (1981): 199–206.

[8] Armstrong, D. M. *A Combinatorial Theory of Possibility*. Cambridge: Cambridge University Press, 1989.

[9] Spinney, Oliver Thomas. "Logical Form and Logical Space in Wittgenstein's *Tractatus*." *Synthese* 200, no. 1 (2022), article 31. doi:10.1007/s11229-022-03470-y.

[10] Lokhorst, Gert-Jan C. "Ontology, Semantics and Philosophy of Mind in Wittgenstein's *Tractatus*: A Formal Reconstruction." *Erkenntnis* 29, no. 1 (1988): 35–75.

[11] Weiss, Max. "Logic in the Tractatus." *The Review of Symbolic Logic* 10, no. 1 (2017): 1–50. doi:10.1017/S1755020316000472.

[12] Walters, Robb. "Saying, Showing, and Structure: A Machine-Checked Reconstruction of the *Tractatus* Ontology." Manuscript under review, 2026. Proof artifact: the `tractatus` Lean 4 repository.

[13] Evans, Richard Prideaux, and Martin Berger. "Cathoristic Logic: A Modal Logic of Incompatible Propositions." arXiv:1411.7158 (2014).

[14] McKinsey, J. C. C. "The Decision Problem for Some Classes of Sentences Without Quantifiers." *The Journal of Symbolic Logic* 8, no. 3 (1943): 61–76.

[15] Horn, Alfred. "On Sentences Which Are True of Direct Unions of Algebras." *The Journal of Symbolic Logic* 16, no. 1 (1951): 14–21.

[16] Benzmüller, Christoph, and Bruno Woltzenlogel Paleo. "Automating Gödel's Ontological Proof of God's Existence with Higher-Order Automated Theorem Provers." In *Proceedings of ECAI 2014*, 93–98. IOS Press, 2014.

[17] de Moura, Leonardo, and Sebastian Ullrich. "The Lean 4 Theorem Prover and Programming Language." In *Automated Deduction – CADE 28*, LNCS 12699, 625–635. Springer, 2021.